# Root Cause Analysis of Ride Completion Decline - Using SQL

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# Key Metrics & Analytical Approaches

## Key Metrics

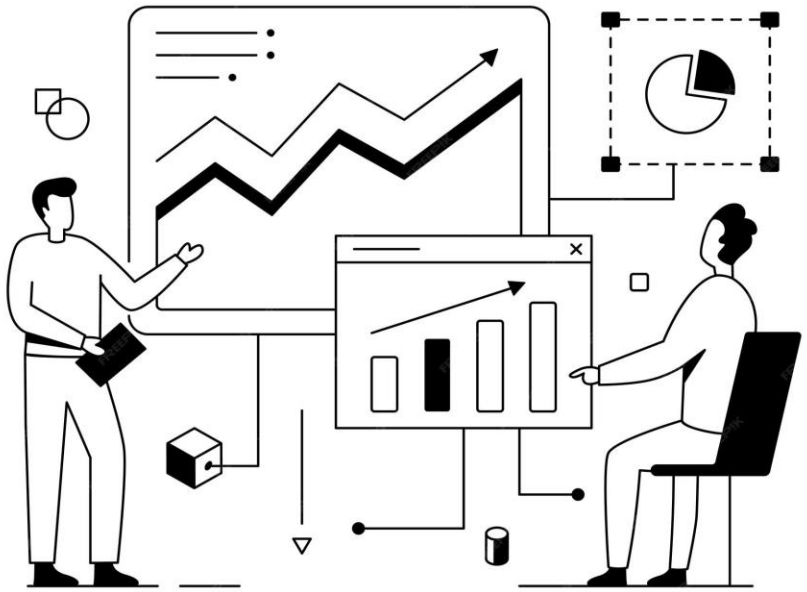
- Ride Completion Rate
- User & Driver Cancellation Rate
- Payment Failure Rate
- Payout Delay %

## Analysis Performed

- Driver Payout Delay Analysis
- Quarterly Ride Completion Trend Analysis (RCA)
- Payment Performance by Payment Mode
- Ride Cancellation Analysis by City and Ride Type
- Ride Completion Performance by Ride Type

# Tools Used

- **SQL –** Data Cleaning, Data Exploration, Data Analysis
- **Power BI** – DAX, Data Visualization
- **Presentation** – Power Point



# Business Problem Statement

The Uber platform experienced a 34% drop in completion rides during Quarter 3, despite ride bookings remaining relatively stable. This indicates that customers are still booking rides, but many of those rides are not being successfully completed.

Such a drop can negatively impact platform revenue, driver utilization, and customer experience, particularly in high-demand urban areas.

The objective of this analysis is to identify the operational and payment-related factors driving the decline in ride completion rates and recommend solutions to improve ride fulfillment.

# Data Cleaning and Preprocessing

1. Identifying Duplicate records:

```
select ride_id, user_id, driver_id, city, ride_type
,booking_time, status, payment_mode, fare_amount, count(*) as cnt
from rides
group by ride_id, user_id, driver_id, city, ride_type
,booking_time, status, payment_mode, fare_amount
having count(*) > 1;
```

2. Removing Duplicate records:

```
delete from rides
where id in (
    select id from
        (select id
        ,row_number() over(partition by ride_id, user_id, driver_id, city, ride_type
        ,booking_time, status, payment_mode, fare_amount) as rn
        from rides) x
where rn > 1);
```

3. Handling Date Format Inconsistencies

```
UPDATE rides
SET booking_time = replace(booking_time, '/', '-');

UPDATE rides
SET booking_time = STR_TO_DATE(booking_time, '%Y-%m-%d %H:%i:%s');

ALTER TABLE rides
MODIFY COLUMN booking_time DATETIME,
MODIFY COLUMN start_time DATETIME,
MODIFY COLUMN end_time DATETIME,
MODIFY COLUMN payout_time DATETIME;
```

# Key Metrics

- Ride Completion Rate

**84.80%**
Ride completion Rate

- Driver Cancellation Rate

**7.80%**
Driver Cancellation Rate

- User Cancellation Rate

**7.40%**
User Cancellation Rate

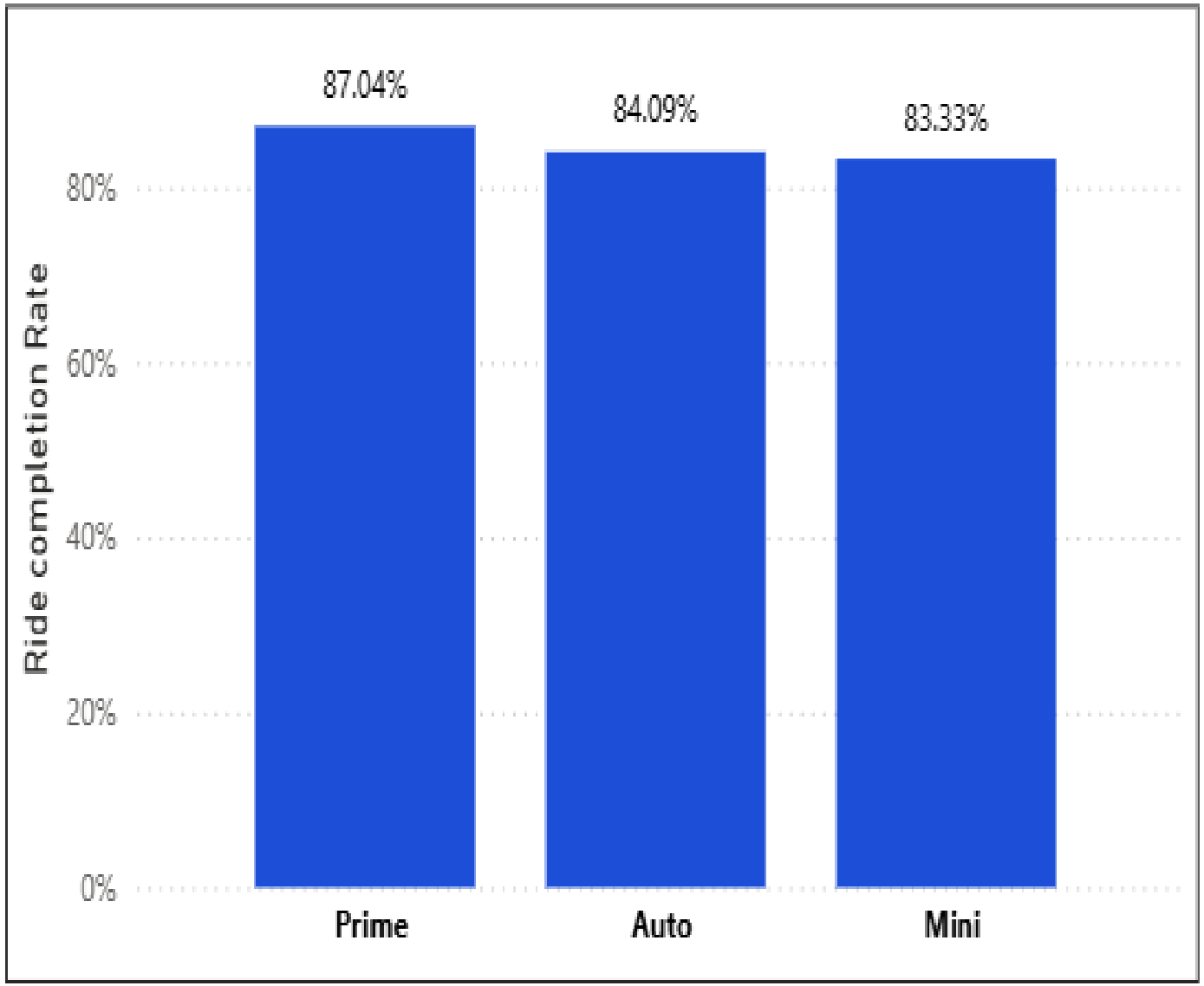
- Payment Failure Rate

**10.14%**
Payout Failure rate

- Payout Delay %

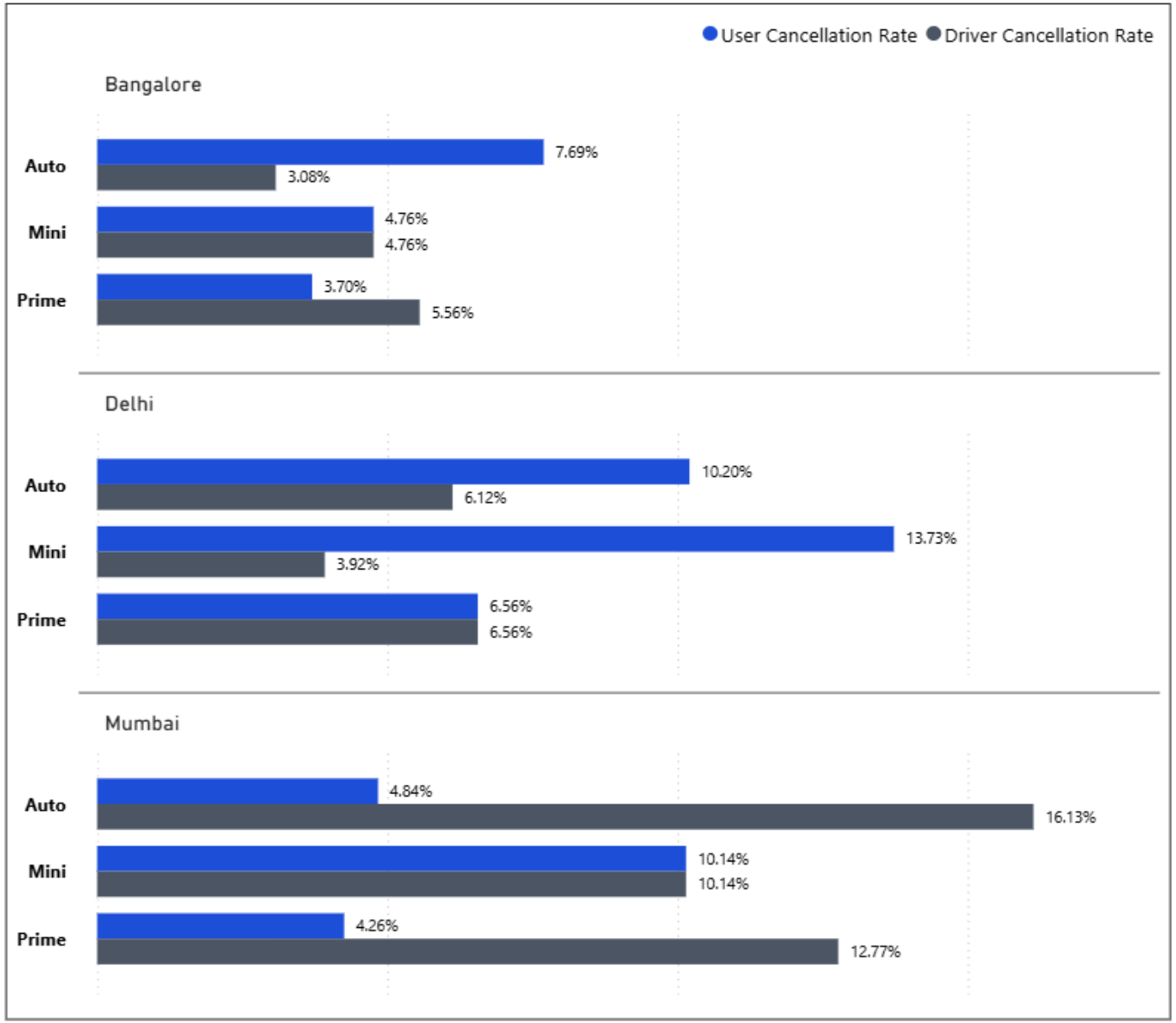
**1.89%**
Payout delay %

# 1. Ride Completion Performance by Ride Type



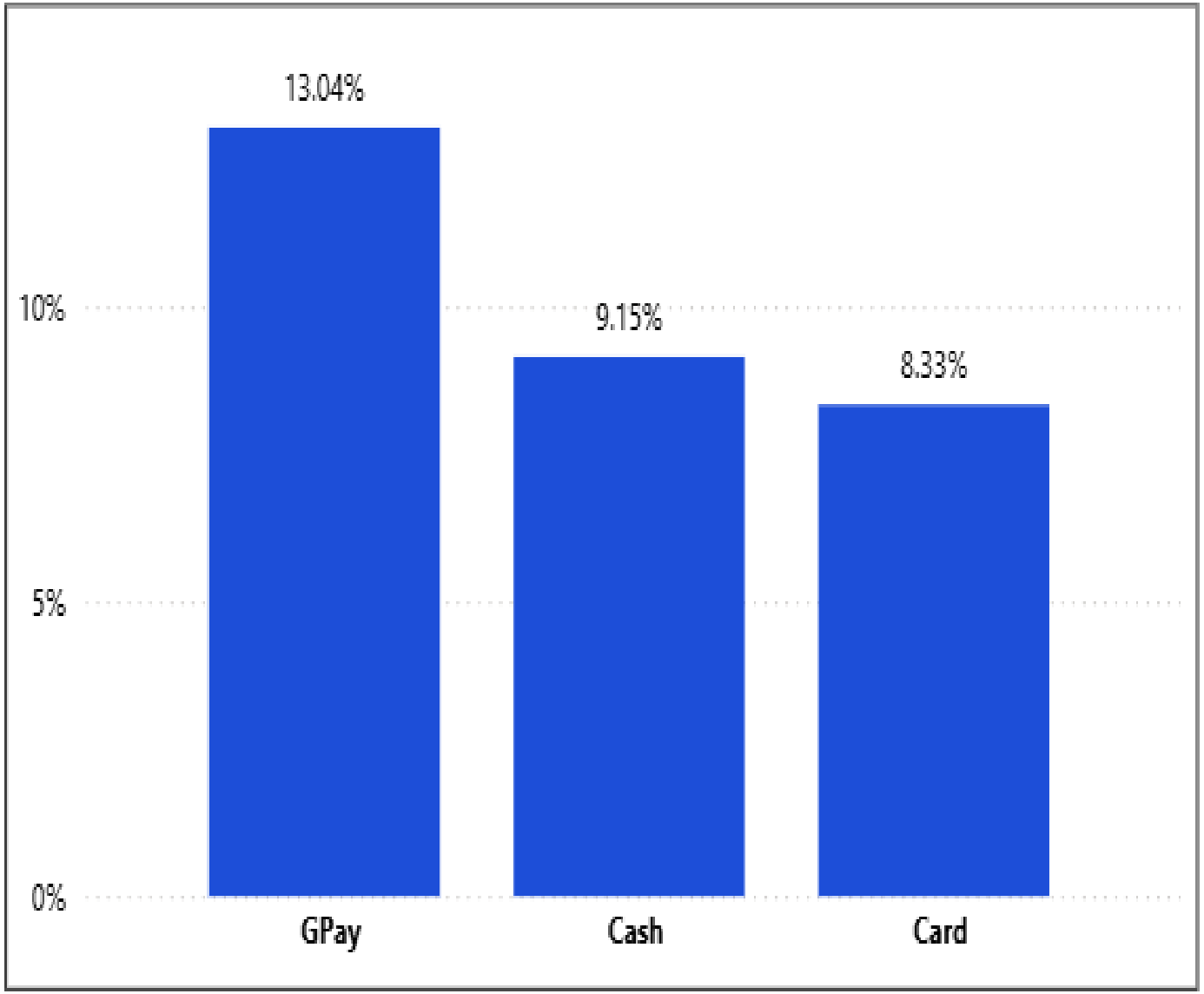

- Prime Rides have the highest completion rate 87%
- Mini Rides have the lowest completion rate 83%
- The difference across ride types is minimal, suggesting consistent fulfillment across services.

# 2. User vs Driver Cancellation Rate by City and Ride Type



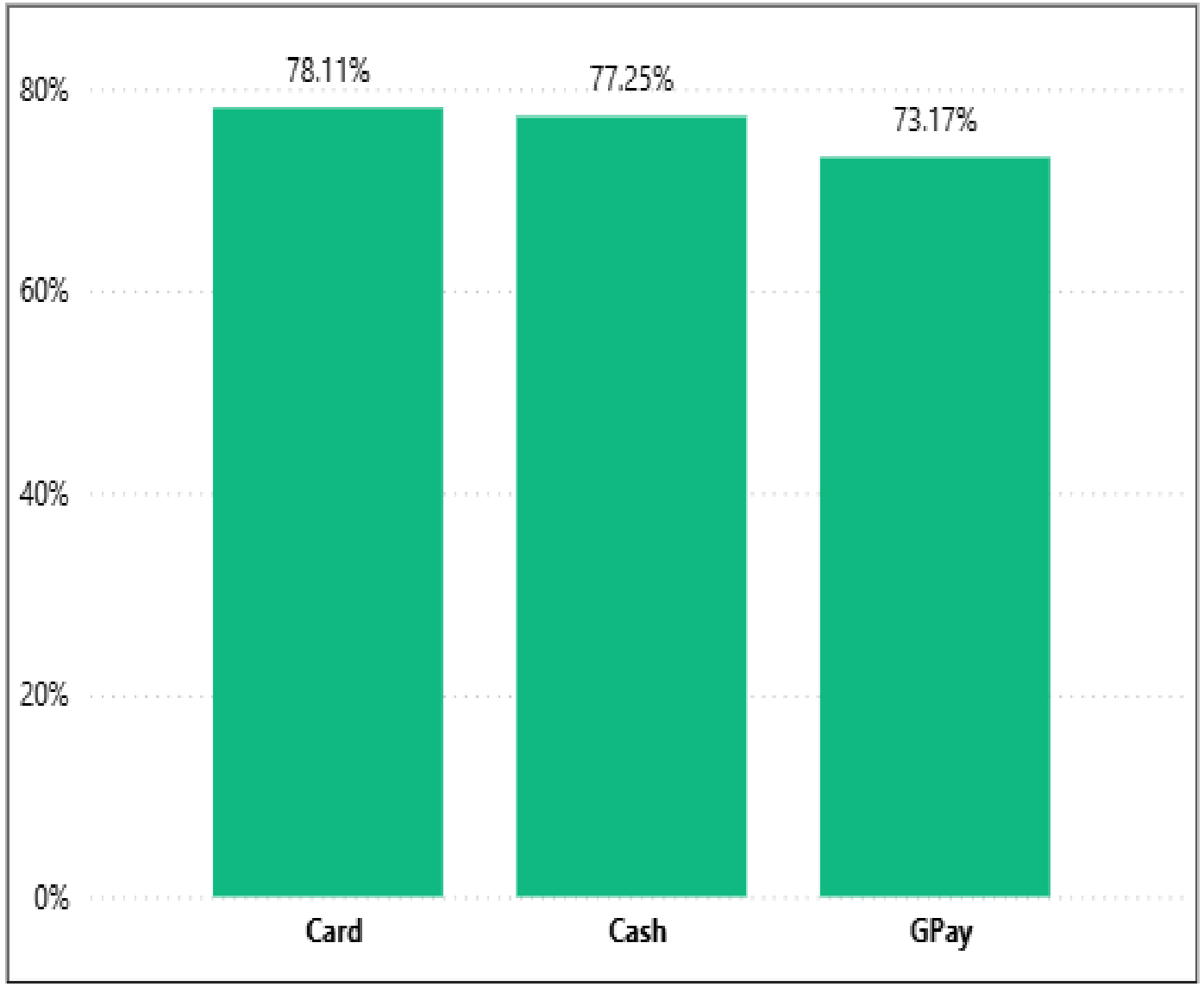

- Driver cancellation are significantly higher in Mumbai, especially in Auto (16.13%) and Prime rides (12.77%).
- Delhi shows higher user cancellations in Mini (13.73%) and Auto rides (10.20%), indicating possible price sensitivity or longer wait time.
- Bangalore has relatively balanced cancellation variability across cities.

# 3. Payment Failure Rate by Payment Mode




- GPay shows the highest payment failure rate (13.04%), indicating bank processing issues.
- Card payments show the lowest failure rate (8.33%), suggesting most reliable payment method

# 4. Payment Success Rate by Payment Mode




- Card payments show the highest payment success rate (78.11%), indicating relatively reliable transaction processing.
- Cash payments also maintain a high success rate (77.25%), slightly lower than card but still stable.
- GPay has the lowest success rate (73.17%), suggesting higher payment failures compared to other modes.

# 5. Driver Payout Delay Analysis

OUTPUT

| payment_status | payout_status | no_of_drivers | avg_payout_delay_days |
|---|---|---|---|
| success | processed | 746 | 0.97 |
| failed | pending | 86 | still delay |
| success | delayed | 16 | 4.98 |

```
with cte as(
select driver_id, ride_type, status as ride_status, payment_status, payout_status
,(timestampdiff(minute, end_time, payout_time)/60) as total_payout_hr
from rides r
inner join payments p
on r.ride_id = p.ride_id
where status = 'completed'
),
cte_2 as
(select driver_id, ride_type, ride_status, payment_status, payout_status
, round(total_payout_hr/ 24,2) as payout_delay_days
from cte)
select payment_status, payout_status, count(*) as no_of_drivers
, coalesce(round(avg(payout_delay_days),2),'still delay') as avg_payout_delay_days
from cte_2
group by payment_status, payout_status;
```

- Most driver payout are processed within 24 hours of ride completion.
- 16 Drivers experienced payout delays (5 days)
- 86 Drivers payouts are still pending, indicating unresolved payment processing issues.
- Payout delays can negatively impact

# 6. Quarterly Ride Completion Trend Analysis (RCA)

OUTPUT

| | quarter | drop_rise_percent |
|---|---|---|
| ▶ | 1 | 0.53 |
| | 2 | 45.57 |
| | 3 | -34.24 |
| | 4 | 5.75 |

```
with summary as
(select month(booking_time) as month
, quarter(booking_time) as quarter
, count(*) as total_request
,round(sum(case when status = 'completed' then 1 else 0 end) * 100/ count(*),2) as completion_rate
,(count(*) * round(sum(case when status = 'completed' then 1 else 0 end) / count(*),2)) as completed_rides
from rides
group by quarter(booking_time), month(booking_time)
order by quarter(booking_time), month(booking_time)),
quarter_bounds as
(select quarter
,min(month) as min_month
,max(month) as max_month
from summary
group by quarter)
select qb.quarter
,round(((max_r.completed_rides - min_r.completed_rides) *100 / min_r.completed_rides),2) as drop_rise_percent
from quarter_bounds qb
join summary min_r
on min_r.quarter = qb.quarter and min_r.month = qb.min_month
join summary max_r
on max_r.quarter = qb.quarter and max_r.month = qb.max_month;
```

- Bookings remained stable in Q3 but completion rides dropped by 34.24%, confirming the problem is in fulfillment not demand.
- GPay, the most used digital payment mode, carried the highest failure rate at 13.04%, directly disrupting ride completion.
- Delay in Driver payout caused drivers to cancel or avoid rides, compounding to completion drop.
- Q4 recovery (+5.75%) suggests the issue is operational and fixable, not a demand problem.

# Actionable Recommendation

- GPay has the highest failure rate at 13.04%, coordinate with GPay's technical team to identify and resolve the root cause before it causes ride cancellations.
- Introduce a fallback payment option: if GPay fails, automatically suggest Cash or Card to prevent ride cancellation.
- Ensure drivers are paid within 24 hours of ride completion to reduce dissatisfaction and prevent drivers from bypassing the app.
- Introduce transparent fare breakdowns in the driver app so drivers understand exactly what they earn per ride, reducing distrust driven cancellations.
- Offer completion bonuses during Q3 (the weakest quarter) to incentivize drivers to fulfil rides despite payment uncertainties.
- Investigate Delhi's Mini and Auto cancellations (13.73%, 10.20%) by user, run price sensitivity tests or reduce estimated wait time display to manage expectations.

# THANK YOU